# TEJ GANGUPANTULA

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## EDUCATION

**University of California, Santa Barbara** December 2026
*B.S. Statistics and Data Science* *Santa Barbara, CA*

- **GPA:** 3.52
- **Relevant Courses:** Statistical Machine Learning, Regression Analysis, Time Series Analysis, Survival Analysis, Stochastic Processes, Database Management Systems, Data Analysis and Inference, Data Science with R/SQL, Probability and Statistics, Calculus I-III, Linear Algebra, Differential Equations

## SKILLS

**AI/Agents:** Claude Code, Codex, LangGraph, LangChain, MCP (TS SDK), Pydantic, Zod, Gemini/Vertex AI, Anthropic API, OpenAI API
**Languages:** Python, R, TypeScript/Node, JavaScript, SQL, Java
**ML/Data:** Scikit-learn, TensorFlow, spaCy/medspaCy, pandas, NumPy, Matplotlib, Seaborn
**Infra/Tools:** Google Cloud Platform, Twilio, React, Docker, FastAPI, PostgreSQL, Git, Streamlit, Jupyter

## PROFESSIONAL EXPERIENCE

**AI Engineer Intern** June 2025 - September 2026
*Agentman (Three Internship Terms: Summer 2025, Spring & Summer 2026)* *San Francisco, CA*

- Built and shipped the MCP tool layer connecting the company's agentic system to electronic health record systems (EHRs): authored the Zod-validated Cross-EHR tool contract and delivered it across 3 EHRs (28 tools each, normalizing REST and SOAP vendor APIs) in TypeScript/Node on GCP Cloud Run, with OAuth-brokered auth and per-tenant credential handles.
- Designed and shipped an end-to-end insurance remittance (X12 835/ERA) pipeline: retrieval, parsing, and denial classification across multiple clinics in TypeScript/Node, published as a versioned library and exposed to agents as MCP tools; found and fixed parser defects that had been silently dropping 57.5% of service-line rows across 1,229 remittance reports.
- Architected and live-verified the company's first outbound voice agent: Twilio Media Streams into a custom audio bridge into Gemini Live speech-to-speech on Vertex AI, with sub-second response latency and barge-in, verified on live two-way calls; authored the architecture spec and build-vs-buy analysis.
- Shipped an admin audit log end-to-end (Python/Postgres API and React desktop app) for a HIPAA-regulated product with role-based access, PHI redaction, and encryption at rest, plus a LangGraph-based chat assistant grounded in the audit trail.
- Wired EHR data sources, third-party healthcare APIs, and vendor-hosted MCP servers into the agents' tool layer via HTTP and MCP tool integrations, with Pydantic schema validation on every tool call for reliability, reducing eligibility verification from 50 to 7 hrs/month and per-check cost from ~$6.72 to $0.50.
- Ran an agentic delivery workflow (Claude Code, Codex, parallel Conductor worktrees) and codified it in org-wide Claude skills I authored (gitflow, EHR-server playbook, deploy verification, code-review subagent); shipped PRs across 8 repos through develop-to-main-to-tag CI/CD on GitHub Actions and Cloud Build into staging and production Cloud Run, with 12 versioned library releases to GitHub Packages.

**Machine Learning Intern** July 2024 - September 2024
*FusionCare* *Davis, CA*

- Built a Python-based clinical text processing pipeline to ingest, clean, and structure de-identified patient notes, applying NLP-based PII masking (spaCy/medspaCy) to safely prepare data for downstream analysis.
- Integrated OpenAI GPT models via LangChain with structured prompting and output validation to convert unstructured clinical notes into JSON-formatted visit summaries, prototyping outputs using Streamlit, improving care plan evaluations and saving clinicians 4 to 5 hours weekly.

## PROJECTS

**Prior Authorization Voice Agent** | *Python, LangGraph, LangChain, FastAPI, PostgreSQL* May 2026 - June 2026

- Built a real-time voice agent (STT → LangGraph state machine → TTS) that answers prior-authorization status questions over a browser interface, routing callers across four normalized outcomes via LangGraph's conditional edges rather than an open-ended ReAct loop.
- Integrated the ElevenLabs API for real-time speech-to-text and text-to-speech, and designed three consent-gated tools (PA lookup, appeal ticket, escalation ticket) backed by a PostgresSaver checkpointer for durable per-call state that survives process restarts.

**NBA Finals Outcome Prediction Analysis** | *Python, Scikit-learn, NumPy, pandas* April 2025 - June 2025

- Developed a Random Forest classification model in Python to simulate and predict 2025 NBA Finals outcomes, achieving 62% test accuracy using team-level regular season and playoff metrics.
- Engineered a Monte Carlo simulation to estimate championship probabilities over a best-of-seven series, predicting a 75.5% chance of the Oklahoma City Thunder winning against the Indiana Pacers (24.5%).